Somatic Polygenic Risk Scoring:

A Novel Framework for Metastasis Risk Stratification

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Abstract

Metastasis is the primary cause of cancer mortality, yet existing genomic biomarkers such as tumor mutational burden (TMB) and microsatellite instability (MSI) primarily inform immunotherapy decisions rather than stratify metastasis risk. This study proposes a novel framework for a somatic polygenic risk score (sPRS), which aggregates weighted somatic tumor variants (gene x variant) to stratify metastasis risk. This is similar to a germline polygenic risk score (PRS) that quantifies disease risk. Weights were computed using MSK-CHORD 2024 data (n = 21,475) via L1-regularized logistic regression with selective penalization, keeping age, sex, race, and cancer type unpenalized. Bootstrap stability analysis across 1,000 samples identified 13 gene x variants that consistently contributed to metastasis risk. sPRS was significantly associated with metastatic status in both unadjusted (AUC = 0.549) and adjusted (AUC = 0.605) models ($p < 0.001$). sPRS also significantly outperformed TMB in discriminative ability in both unadjusted and adjusted models (DeLong's $p < 0.001$), and showed meaningful risk stratification across tertiles. All pairwise comparisons were significant after Bonferroni correction ($\alpha = 0.017$). This study demonstrated that weighted aggregation of tumor genetic variants offers a promising and clinically accessible method for stratifying metastasis risk.

Introduction

While cancer sits at the number two leading cause of death in the US, metastasis is the major driver of these cancer deaths, often cited as accounting for 90% of cancer mortality.[1] With

current advancements in modeling through techniques such as support vector machines, decision trees, and neural networks, it has become easier to predict metastasis. Enriched genomic data, liquid biopsies, and radiomics are now being incorporated into these models, further improving their performance.[2] However, this is a heavy data collection and infrastructure burden on the clinical system, and ultimately the patient, who is already battling a primary cancer.

As oncogenomics continues to expand and advance, measures such as tumor mutational burden (TMB) and microsatellite instability (MSI) have been developed as predictive biomarkers that characterize various aspects of tumor mutational biology. These values are primarily used to inform immunotherapy treatment decisions, rather than directly predicting metastasis risk.[3] Additionally, TMB is the unweighted count of non-synonymous mutations present in the tumor per megabase. There is no distinction between mutation *type* (missense, nonsense, splice site, etc.) or any weighting factor to indicate the varied effect of those mutations.[3] While TMB and MSI characterize the tumor mutational landscape, neither produces a composite risk score that weights variants individually towards a specific clinical outcome (i.e. metastasis).

Another common measure constructed from genomic data is a polygenic risk score (PRS). A PRS is an aggregated measure of risk for some biological event.[4] For example, these risk scores are commonly used to identify patients who may benefit from earlier cancer screening, due to the cumulative effect of common genetic variants (SNPs) that together indicate a higher likelihood of a specific cancer from germline data.[4] A PRS, however, does not represent the *tumor* genomic landscape.

This study proposes a novel framework for a somatic polygenic risk score (sPRS). Similar to a PRS, an sPRS will aggregate the *weighted* burden of genetic variants found in the

tumor genomic landscape to stratify metastasis risk. This measure can then be used to inform clinical decisions to mitigate this risk.

Methods

Data and Study Population

Data came from the Memorial Sloan Kettering clinicogenomic, harmonized oncologic real-world dataset (MSK-CHORD: publicly available through cBioPortal for Cancer Genomics), from which three data files were used: clinical patient data (age, race, and gender), clinical sample data (metastasis status, cancer type, MSI score, and non-synonymous TMB), and somatic mutation data (gene and variant classification).[5] Patients were sequenced using one of the five MSK IMPACT targeted gene panels: IMPACT341, IMPACT410, IMPACT468, IMPACT505, or IMPACT-HEME-400.

The study population consisted of all patients in the MSK-CHORD 2024 database that met the following criteria: unique sample type (excluding patients who had *both* a primary and metastasized tumor sequenced to avoid spillover), sample types specifically identified as primary or metastasis (excluding unknown or local recurrence samples), full race information (excluding missing, unknown, or refused race information), and genes sequenced across all five IMPACT panels with a mutation rate greater than 1%. Complete case analysis was applied to the remaining cohort.

Gene Panel Harmonization

Because patients were sequenced using five different gene panels of varying size, genes that were common between all five panels were retained for analysis. This was done to prevent misclassification of a mutation as “absent” when that gene may not have been sequenced. Gene naming conventions were also verified to ensure consistency across panels.

Data Splitting

The final cohort was partitioned into a training set and a test set using a 70/30 stratified split. Stratification was done considering both cancer type (Colorectal, Non-Small Cell Lung, Breast, Pancreatic, and Prostate) and binary metastatic status to ensure representative proportions of both variables across the splits. All model development was done using the training set, and sPRS construction and evaluation were performed using the test set.

sPRS Construction

The sPRS was calculated using equation (1), where $\hat{\beta}_j^{LASSO}$ is the weight for gene x variant j, and x_{ij} indicates whether patient i carries gene x variant j.

$$sPRS_i = \sum_{j=1}^{J} \hat{\beta}_j^{LASSO} * x_{ij} \tag{1}$$

Weights Extraction

The weights were constructed via L1-regularized logistic regression, with the model specified in equation (2). Note that only the gene x variant indicator, x_{ij} , was penalized, leaving age, sex, race, and cancer type unpenalized in the model. This was done to ensure that the L1-regularization procedure selected exclusively among the gene x variant features, reflecting the independent contribution to metastasis risk adjusted for established clinical and demographic confounders.

$$logit(\pi_i) = \beta_0 + \sum_{j=1}^{J} \beta_j^{LASSO} x_{ij} + \beta_a age_i + \beta_s sex_i + \beta_r race_i + \beta_c cancer_i \tag{2}$$

Equation (3) shows the L1-regularization applied to the gene x variant. The regularization parameter, λ , was selected via 10-fold cross validation over a grid of 100 candidate values, using the receiver operating characteristic curve (AUC) as the selection criterion. The λ that minimized

cross-validated loss (λ_{min}) was selected, as the primary goal was predictive performance rather than maximum sparsity.

$$\hat{\beta}^{LASSO} = \arg\min_{\beta}\left[-\, l(\beta) + \lambda \sum_{j=1}^{J} w_j \beta_j\right] \tag{3}$$

where $w_j = \{0 \; if \; j \; is \; demographic, \; 1 \; if \; j \; is \; a \; gene \; variant\}$

sPRS Evaluation

The constructed sPRS scores for each patient in the test set were evaluated on four factors: association with metastatic status, discriminative ability, performance compared to TMB and MSI, and risk stratification ability. sPRS was not designed to be the sole predictor of metastasis, but rather to be a component in the broader clinical decision making process, which is consistent with how germline PRS is clinically used.

Binomial logistic regression using sPRS as the predictor for metastasis was used to evaluate the association between sPRS and metastasis. Two models were fit: an unadjusted model with sPRS as the sole predictor, and an adjusted model that included age, sex, race, and cancer type as covariates. Statistical significance was assessed via a likelihood ratio test (LRT), and reported alongside odds ratios and 95% confidence intervals.

The AUC was used to assess the discriminative ability of the sPRS, both using the adjusted and unadjusted models, and reflects the probability that a randomly selected metastatic patient received a higher predicted probability than a randomly selected primary tumor patient.

Performance against TMB and MSI was assessed by fitting separate unadjusted and adjusted logistic regression models for TMB and MSI with the same covariate set used in the sPRS adjusted model above, and comparing AUCs using DeLong's test for correlated ROC curves with a significance level of $\alpha = 0.05$. MSI was empirically observed to be inversely

associated with metastatic status, and thus the ROC curve for MSI was constructed in the decreasing direction to correctly compare sPRS, TMB, and MSI.

Patients in the test set were stratified into tertiles of sPRS (low, medium, and high), and the metastatic rate within each of these groups was computed with 95% confidence intervals. Pairwise comparisons were assessed using chi-square tests with Bonferroni correction at an adjusted significance level of $\alpha = \frac{0.05}{3} = 0.017$.

Results

The final cohort consisted of 21,475 unique patients, with 15,031 in the training split and 6,444 in the test split. **Table 1** shows the demographics of the full cohort and by metastasis status. Note that most demographic information summary statistics amongst primary and metastatic patients are quite similar, with the exception of biologically motivated processes, such as colorectal cancer metastatic status, breast cancer metastatic status, and highest stage recorded.

Table1. Cohort Characteristics

Characteristic	Overall N = 21,475[1]	Metastasis N = 7,704	Primary N = 13,771	p-value[2]
Age	65 (13)	65 (12)	66 (13)	<0.001
Gender				<0.001
Female	12,055 (56%)	4,572 (59%)	7,483 (54%)	
Male	9,420 (44%)	3,132 (41%)	6,288 (46%)	
Race				0.017
White	17,476 (81%)	6,199 (80%)	11,277 (82%)	
Asian-Far East/Indian Subcont	1,759 (8.2%)	646 (8.4%)	1,113 (8.1%)	
Black Or African American	1,593 (7.4%)	636 (8.3%)	957 (6.9%)	
Other	610 (2.8%)	210 (2.7%)	400 (2.9%)	
Native American-Am Ind/Alaska	25 (0.1%)	8 (0.1%)	17 (0.1%)	
Native Hawaiian Or Pacific Isl	12 (<0.1%)	5 (<0.1%)	7 (<0.1%)	
Cancer Type				<0.001
Non-Small Cell Lung Cancer	6,528 (30%)	2,354 (31%)	4,174 (30%)	
Colorectal Cancer	5,143 (24%)	1,327 (17%)	3,816 (28%)	
Breast Cancer	4,612 (21%)	2,145 (28%)	2,467 (18%)	
Pancreatic Cancer	2,761 (13%)	992 (13%)	1,769 (13%)	
Prostate Cancer	2,431 (11%)	886 (12%)	1,545 (11%)	
Highest Stage Recorded				<0.001
Stage 1-3	13,041 (61%)	3,541 (46%)	9,500 (69%)	
Stage 4	8,434 (39%)	4,163 (54%)	4,271 (31%)	
TMB	8 (17)	7 (10)	8 (19)	<0.001
MSI Score	1.51 (5.16)	1.16 (3.13)	1.70 (5.99)	<0.001
Missing	79 (0.4%)	28 (0.4%)	51 (0.4%)	

[1] Mean (SD); n (%)

[2] Wilcoxon rank sum test; Pearson's Chi-squared test except for Race which used Fisher's Exact Test with simulated p-value due to small cell counts

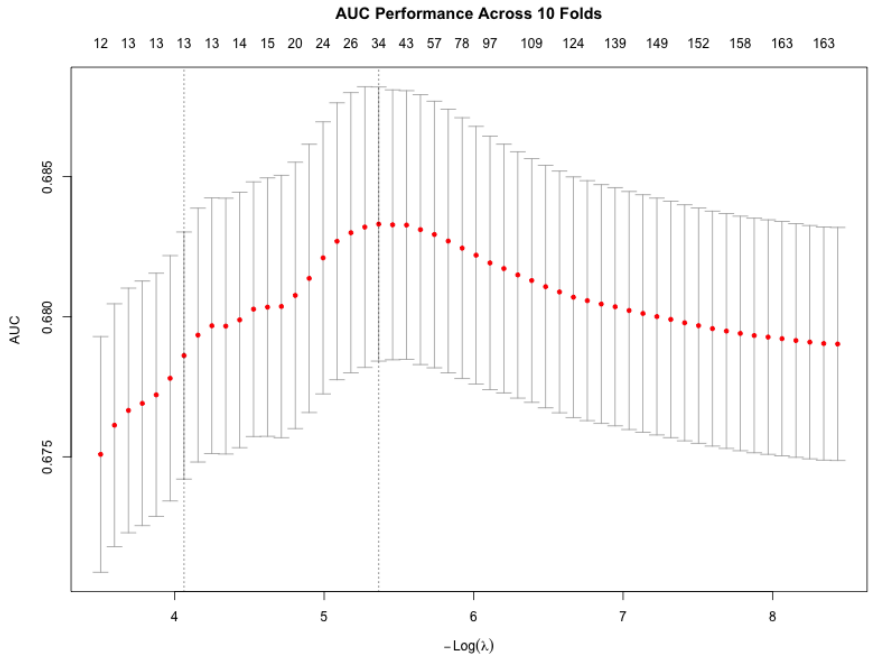

Figure 1. AUC performance across 10 folds to select gene x variants

The 10-fold cross validation of the L1-regularized logistic regression selected 34 total predictors, of which 22 were gene x variant predictors (**Figure 1**).

After bootstrapping across 1,000 samples with the regularization parameter set to the minimum lambda identified in **Figure 1**, 13 of those gene x variants survived 70% of the time or more (**Table 2**).

Table 2. Surviving gene x variants after bootstrap stability analysis

Gene	Mutation Type	sPRS Weight
ESR1	Missense	0.845
ARID1B	Missense	0.184
KEAP1	Missense	0.140
ARID1A	Nonsense	0.129
AR	Missense	0.128
ERBB4	Missense	0.107
TP53	Missense	0.097
TP53	Splice Site	0.079
TP53	Nonsense	0.026
PIK3CA	Missense	-0.067
KMT2C	Missense	-0.113
FBXW7	Missense	-0.191
RNF43	Frame Shift Deletion	-0.390

After construction of the sPRS score as shown in (1) on the test set, two binomial logistic regression models were built with sPRS as the predictor of interest: unadjusted and adjusted. **Table 3** shows that sPRS is associated with metastatic status in both unadjusted and adjusted models ($p < 0.001$). Additionally, the AUC of 0.549 for the unadjusted and 0.605 for the adjusted show that sPRS has a moderate discriminative ability with respect to metastatic status.

Table 3. Association between sPRS and metastatic status

Model	Odds Ratio	95% CI	LRT p-value[1]	AUC
Unadjusted	3.889	(2.799, 5.404)	1.10e-16	0.549
Adjusted[2]	3.720	(2.645, 5.233)	6.90e-41	0.605

[1] LRT p-value tests the null hypothesis that sPRS does not improve model fit over the intercept-only model.
[2] Adjusted model includes age, sex, race, and cancer type as covariates.

AUC comparisons between sPRS, TMB, and MSI are shown in **Table 4**. sPRS significantly discriminated metastatic status better than TMB in both the unadjusted and adjusted models (unadjusted: AUC sPRS = 0.549, AUC TMB = 0.477, DeLong's $p < 0.001$; adjusted: AUC sPRS = 0.605, AUC TMB = 0.599, DeLong's $p <0.001$). MSI demonstrated slightly higher discrimination than sPRS in the unadjusted model, but slightly lower in the adjusted model (unadjusted: AUC sPRS = 0.549, AUC MSI = 0.600, DeLong's $p < 0.001$; adjusted: AUC sPRS = 0.605, AUC MSI = 0.598, DeLong's $p <0.001$). MSI was empirically observed to be inversely associated with metastatic status in this cohort, which is consistent with the known biology of MSI-high tumors which are associated with increased immune infiltration and more favorable clinical trajectories across many cancer types.[6] Although MSI showed a higher AUC in the unadjusted model, MSI testing has known limitations in non-colorectal cancers, and is not a universally informative measure across cancer types. sPRS, which was derived from solely

somatic mutation profiles available from routine tumor sequencing, offers a complementary usage where MSI is uninformative or unavailable.

Table 4. AUC comparison across sPRS, TMB, and MSI

Measure	AUC (Unadjusted)	AUC (Adjusted)[1]	DeLong p-value vs sPRS (Unadjusted)[2]	DeLong p-value vs sPRS (Adjusted)
sPRS	0.549	0.605	-	-
TMB	0.477	0.599	1.18e-10	4.00e-02
MSI[3]	0.600	0.598	8.98e-07	4.62e-01

[1] Adjusted models include age, sex, race, and cancer type as covariates.

[2] DeLong's test for correlated ROC curves. Significance threshold $\alpha = 0.05$.

[3] MSI ROC curve constructed in the decreasing direction due to empirically observed inverse association with metastatic status.

Figure 2 shows the sPRS split into tertiles (low, medium, and high), and the average metastatic rate and 95% confidence interval bars within each of those groups.

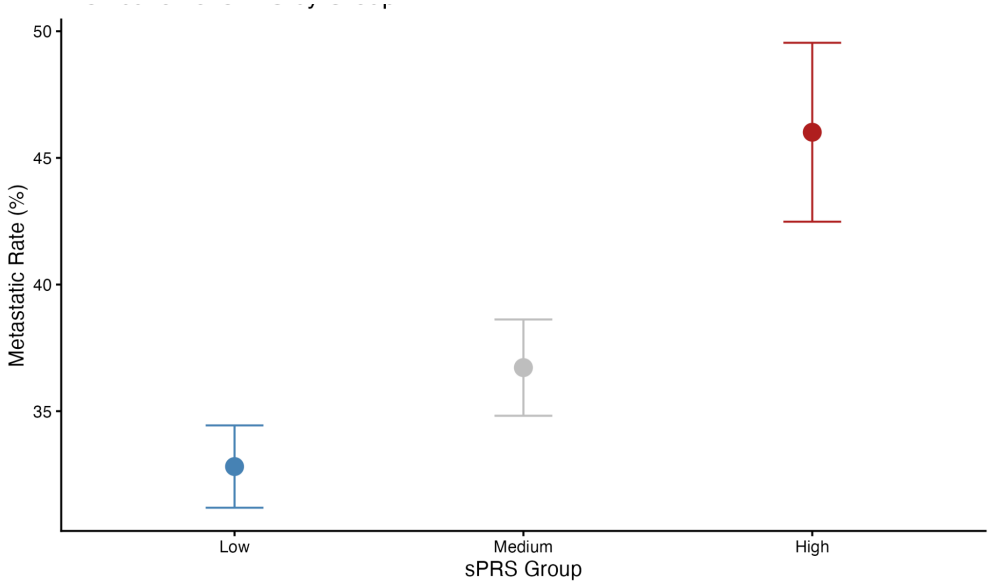

Figure 2. Distribution of sPRS by tertile group

The overall chi-squared test was significant ($p < 0.001$). All pairwise comparisons (Low vs. Medium, Low vs. High, Medium vs. High) were statistically significant after Bonferroni correction ($\alpha = 0.017$), which shows that the intended purpose for the sPRS was achieved.

Similar to a PRS, sPRS can be grouped into distinct groups with regards to metastatic risk to characterize patients based on their tumor mutational biology.

Discussion

This study proposed and validated a novel sPRS framework for metastasis risk stratification using tumor somatic mutation profiles from MSK-CHORD. sPRS was significantly associated with metastatic status in both unadjusted and adjusted models, outperformed TMB in discriminative ability in both unadjusted and adjusted settings, and demonstrated meaningful risk stratification across tertiles.

The sPRS AUCs of 0.549 and 0.605 for the unadjusted and adjusted models respectively are consistent with expectations for a score constructed in this manner. Germline PRS scores for complex traits achieve AUCs ranging from 0.55 to 0.65, and are still considered clinically meaningful.[7] Both sPRS and PRS are intended to contribute to a broader risk assessment, rather than the sole predictor.[7] The improvement in AUC from unadjusted to adjusted suggests sPRS captures gene x variant information that is complementary to demographic factors.

Additionally, the significant outperformance of sPRS compared to TMB (DeLong's $p < 0.001$) supports the argument that weighting somatic variants by their independent contribution adds value over an unweighted simple mutation count. TMB treats all non-synonymous mutations as equal, regardless of type or functional consequence, while sPRS differentiates between these gene x variants.

The MSI comparison, however, showed higher discrimination than sPRS in the unadjusted models, but lower discrimination in the adjusted models. This suggests that MSI's unadjusted performance is in part due to its known association with specific cancer types,

especially colorectal cancer.[8] In addition, the inverse association observed between MSI and metastatic status is consistent with known biology of tumors with high MSI.[6]

This study has several limitations. First, metastatic status was defined cross-sectionally based on sample type at time of sequencing, which does not capture longitudinal metastatic progression. A patient with a primary tumor sample may have metastasized without record of it in MSK-CHORD. Second, complete case analysis was used which potentially introduces selection bias if the excluded patients were systematically different from the study cohort. Third, MSK-CHORD is a single tertiary cancer center, which makes it harder to generalize these results to a broader population. Fourth, sPRS construction was limited to genes common across the five IMPACT panels, which may have missed biologically relevant mutations that larger panels sequenced. Finally, the sPRS was validated within MSK-CHORD itself, and thus requires external validation in an independent cohort.

Further work should focus on external validation of these results, extending to longitudinal validation to assess whether sPRS at primary diagnosis predicts the development of metastasis later on, and creating cancer-specific sPRS scores that potentially offer higher discriminative ability than the pan-cancer approach taken here. A natural extension of this proposed framework is incorporating sPRS with other biomarker data coming from imaging and clinical staging to create a composite risk model.

Conclusion

This study proposed a novel framework for somatic polygenic risk scoring from tumor genetic data. The sPRS demonstrates significant association with metastasis, better performance than traditional tumor mutation measures such as TMB, and meaningful tertile-level risk

stratification. These findings support the potential usefulness of sPRS as a component in the broader clinical decision making process in precision oncology.

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